

16.07: Dynamics

Fall 2026

Course Description

This course is about modeling and simulating mechanical systems, starting with a review of basic Newtonian mechanics of point-mass particles, then covering rigid-body and multi-body systems. Applications to aircraft, spacecraft, and robotic systems will be emphasized.

Prerequisites: Officially – (16.001 or 16.002) and (16.003 or 16.004). Practically – basic mechanics at the level of 8.01, calculus and vectors/matrices at the level of 18.02, and prior experience with a high-level programming language like Python or MATLAB (we'll use MATLAB).

Instructors

Lecturer: Zac Manchester	Email: zacm@mit.edu	Office: 31-243
TA: John Zhang	Email: jzhang3@mit.edu	

Logistics

- Lectures will be held **Tuesdays and Thursdays 11:00–12:30 in 35-225**. Lectures will also be recorded for later viewing.
- Recitations will be held **Fridays 3:00–4:00 in 35-225**. These will primarily be used for office hours, but three of these times will be used for labs.
- Office hours will be **Tuesdays 4:00–5:00 in 31-201M** (location may change in the future).
- Homework assignments will be due **Fridays by 11:59 PM** most weeks.
- There will be a short (5 minute) quiz in class corresponding to each homework assignment. Quizzes will be in class on the Tuesday after the corresponding homework deadline.
- There will be an in-class mid-term exam and a cumulative final exam.
- Piazza will be used for general discussion and Q&A outside of class and office hours.

Learning Objectives

By the end of this course, students should be able to do the following:

1. Select appropriate coordinates to describe the motion of particle and multi-body system
2. Formulate dynamics in non-inertial reference frames
3. Identify and exploit conserved quantities like energy and momentum
4. Derive equations of motion for multi-body systems using the Euler-Lagrange equation
5. Model the orbital dynamics of a spacecraft as central-force motion of a particle
6. Use Euler's equation to model and analyze the attitude dynamics of a rigid body and apply this to aircraft and spacecraft

Learning Resources

There is no required textbook for this course. Video recordings of lectures and lecture notes will be posted online. The following books are good references for the topics discussed in the course that you may want to refer to:

1. J. L. Meriam, L. G. Kraige, and J. N. Bolton, Engineering Mechanics: Dynamics, 9th ed. Hoboken, NJ, USA: Wiley, 2020.
2. L. D. Landau and E. M. Lifshitz, Mechanics, 3rd ed. Oxford, U.K.: Butterworth-Heinemann, 1976.
3. H. Goldstein, C. Poole, and J. Safko, Classical Mechanics, 3rd ed. San Francisco, CA, USA: Addison Wesley, 2002.
4. K. M. Lynch and F. C. Park, Modern Robotics: Mechanics, Planning, and Control. Cambridge, U.K.: Cambridge University Press, 2017.

Assignments

- Students will be asked to complete homework assignments roughly weekly that include both an analytical question and a numerical simulation question to be completed in MATLAB.
- There will be three labs throughout the semester during the Friday recitation time. On lab weeks, homework assignments will include analysis of experiment data.

Quizzes and Exams

- Short quizzes will be given approximately weekly in class. These are intended to take no more than 5 minutes to complete.
- There will be an in-class mid-term exam and a cumulative final exam.

Grading

Grading will be based on:

- 25% Homework and Labs
- 15% Quizzes
- 25% Mid-Term Exam
- 30% Final Exam
- 5% Participation

Students can earn a full participation grade through any combination of office hours, Piazza discussions, recitation attendance, and by offering constructive feedback about the course to the instructors.

Course Policies

Late Assignments: Students are allowed a budget of six late days for turning in homework with no penalty throughout the semester. They may be used together on one assignment, or separately across multiple assignments. Beyond these six days, no other late homework will be accepted.

Quiz Grades: The lowest quiz grade will be dropped from each student's semester cumulative grade. This can also be used to cover an absence.

Collaboration: Students are encouraged to collaborate on labs and homework assignments, but each student must turn in their own assignment by the deadline.

AI Use: Students are permitted to use AI tools like Claude and Gemini for assistance on homework assignments. This should be acknowledged explicitly with a brief statement about how AI was used.

Tentative Schedule

Week	Dates	Topics
1	Sep 10	Course Overview, History, and Review
2	Sep 15 Sep 17	Coordinates and Newtonian Mechanics Review State Space, Numerical Integration, Energy, and Lyapunov Stability
3	Sep 22 Sep 24	Work, d'Alembert's Principle, Euler-Lagrange Equation Manipulator Equation, Mass Matrix
4	Sep 29 Oct 1	Dynamics in Rotating Frames Linear and Angular Momentum, Rigid Bodies
5	Oct 6 Oct 8	Attitude Kinematics, Rotation Matrices Quaternions and Simulation
6	Oct 13 Oct 15	No Class — Monday Schedule Euler's Equation, Equilibria
7	Oct 20 Oct 22	Linearization and Stability, Spinning Spacecraft Newton-Euler Dynamics for Aircraft
8	Oct 27 Oct 29	Mid-Term Review Mid-Term Exam (Zac Out)
9	Nov 3 Nov 5	Intrinsic Coordinates and Frenet-Serret (Zac Out) TBD (Zac Out)
10	Nov 10 Nov 12	Gravitational Potential, The Kepler Problem Orbital Elements, Perturbations, The J2 Effect
11	Nov 17 Nov 19	Orbital Transfers The Circular Restricted Three-Body Problem, Equilibria and Stability
12	Nov 24 Nov 26	3D Multi-Body Dynamics and Simulation No Class — Thanksgiving
13	Dec 1 Dec 3	Constraints, Lagrange Multipliers, and Constraint Stabilization Coulomb Friction and the Maximum Dissipation Principle
14	Dec 8 Dec 10	System Identification and Least-Squares TBD: Schedule Margin